

HTML/CSS Project

Build and Deploy a Complete Portfolio Website

■ CAPSTONE PROJECT

■ LEARNING OUTCOME

By the end of this module you will design, build, and deploy a complete, responsive, accessible portfolio website using only HTML and CSS — applying everything from modules 1-9 — and publish it live with GitHub Pages.

■ Skills You Will Gain

- Plan a 6-section portfolio with wireframes
- Write semantic HTML5 structure for all sections
- Set up CSS custom properties for a design system
- Build hero, about, skills, projects, contact sections
- Make the layout fully responsive with mobile-first CSS
- Publish live with GitHub Pages and a custom domain

■ Your portfolio website is your 24/7 recruiter — it works while you sleep. One good portfolio gets you 3-5x more interview callbacks than no portfolio. Build it once, update continuously.

SECTION 1

Project Planning

1	Define Your Brand Who are you? What role do you want? What makes you different? Write 1 sentence that answers all three.
2	Choose 3-5 Projects Your best work — pick projects that solve real problems, not just tutorials you followed.
3	Wireframe on Paper Sketch each section: what content, what layout, where CTAs. NEVER skip wireframing.
4	Set Up File Structure index.html, style.css, assets/images/, assets/fonts/ — clean from the start.
5	Design Tokens First Define colours, fonts, spacing in CSS variables before writing any component CSS.
6	Build Section by Section Complete each section before moving to next — don't half-build everything.
7	Mobile-First Build Start with mobile styles, then add complexity for larger screens.
8	Deploy Early Push to GitHub Pages on day 1 — deploy often, catch issues early.

SECTION 2

HTML Structure

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <meta name="description" content="Jane Doe –
Frontend Developer specialising in React and CSS"> <meta property="og:title" content="Jane Doe |
Frontend Developer"> <meta property="og:description" content="Building fast, accessible web apps">
<meta property="og:image" content="https://janedoe.dev/og-image.jpg"> <meta property="og:url"
content="https://janedoe.dev"> <title>Jane Doe | Frontend Developer</title> <link rel="stylesheet"
href="style.css"> <link rel="icon" href="favicon.ico"> </head> <body> <!-- SKIP LINK –
accessibility: keyboard users skip to main content --> <a href="#main" class="skip-link">Skip to
main content</a> <header class="site-header" id="top"> <a href="#top" class="logo" aria-label="Back
to top">JD</a> <nav aria-label="Main navigation"> <ul class="nav__list"> <li><a
href="#about">About</a></li> <li><a href="#skills">Skills</a></li> <li><a
href="#projects">Projects</a></li> <li><a href="#contact">Contact</a></li> </ul> </nav> <button
class="nav__toggle" aria-expanded="false" aria-controls="nav-menu" aria-label="Toggle menu">
<span></span><span></span><span></span> </button> </header> <main id="main"> <section class="hero"
aria-label="Introduction"> <div class="hero__content"> <p class="hero__label">Hello, I'm</p> <h1
class="hero__name">Jane Doe</h1> <p class="hero__role">Frontend Developer</p> <p
class="hero__tagline">I build fast, accessible, beautiful web experiences.</p> <div
class="hero__actions"> <a href="#projects" class="btn btn--primary">View My Work</a> <a
href="#contact" class="btn btn--outline">Let's Talk</a> </div> </div> </section> <section
class="about" id="about"> <h2>About Me</h2> <div class="about__grid"> 
```

```

class="about__photo"> <div class="about__text"> <p>I'm a frontend developer with 2 years of
experience building responsive web applications with React, TypeScript, and CSS.</p> <p>When I'm
not coding, I'm contributing to open source and writing about web accessibility on my blog.</p> <a
href="assets/jane-doe-cv.pdf" class="btn btn--primary" download aria-label="Download Jane Doe's CV
PDF">Download CV</a> </div> </div> </section> <section class="skills" id="skills"> <h2>Skills</h2>
<div class="skills__grid"> <article class="skill-card">  <h3>HTML</h3> <p>Semantic, accessible markup</p> </article> <!-- Repeat for
CSS, JavaScript, React, etc. --> </div> </section> <section class="projects" id="projects">
<h2>Projects</h2> <div class="projects__grid"> <article class="project-card">  <div class="project-card__body"> <h3>E-Commerce
Dashboard</h3> <p>Real-time sales analytics built with React and Chart.js.</p> <ul class="tags"
aria-label="Technologies used"> <li>React</li><li>TypeScript</li><li>CSS Grid</li> </ul> <div
class="project-card__links"> <a href="https://demo.janedoe.dev/proj1" target="_blank"
rel="noopener" aria-label="Live demo of E-Commerce Dashboard">Live Demo</a> <a
href="https://github.com/jane/proj1" target="_blank" rel="noopener" aria-label="GitHub code for
E-Commerce Dashboard">GitHub</a> </div> </div> </article> </div> </section> <section
class="contact" id="contact"> <h2>Let's Work Together</h2> <form
action="https://formspree.io/f/YOUR_ID" method="POST"> <div class="form-group"> <label
for="contact-name">Name</label> <input type="text" id="contact-name" name="name" required
autocomplete="name" placeholder="Your name"> </div> <div class="form-group"> <label
for="contact-email">Email</label> <input type="email" id="contact-email" name="email" required
autocomplete="email" placeholder="you@example.com"> </div> <div class="form-group"> <label
for="contact-message">Message</label> <textarea id="contact-message" name="message" rows="5"
required placeholder="Tell me about your project..."></textarea> </div> <button type="submit"
class="btn btn--primary">Send Message</button> </form> </section> </main> <footer
class="site-footer"> <p>&copy; <time datetime="2024">2024</time> Jane Doe. Built with HTML &
CSS.</p> <nav aria-label="Social links"> <a href="https://github.com/jane" target="_blank"
rel="noopener me" aria-label="Jane on GitHub">GitHub</a> <a href="https://linkedin.com/in/jane"
target="_blank" rel="noopener me" aria-label="Jane on LinkedIn">LinkedIn</a> </nav> </footer>
</body> </html>

```

SECTION 3

CSS Framework — Design System

```

/* === 1. DESIGN TOKENS === */ :root { /* Brand colours */ --clr-primary: #E44D26;
--clr-primary-dark: #c43e1c; --clr-bg: #0f0f0f; --clr-surface: #1a1a1a; --clr-text: #f5f5f5;
--clr-text-muted: #a3a3a3; /* Type scale */ --fs-sm: clamp(.875rem, 1.5vw, 1rem); --fs-base:
clamp(1rem, 2vw, 1.125rem); --fs-lg: clamp(1.25rem, 3vw, 1.75rem); --fs-xl: clamp(2rem, 6vw, 4rem);
--fs-2xl: clamp(2.5rem, 8vw, 5.5rem); /* Spacing */ --sp-xs: 8px; --sp-sm: 16px; --sp-md: 24px;
--sp-lg: 48px; --sp-xl: 96px; /* Typography */ --font-body: 'Inter', system-ui, sans-serif;
--font-heading: 'Poppins', system-ui, sans-serif; /* Effects */ --shadow-sm: 0 1px 3px
rgba(0,0,0,.3); --shadow-md: 0 4px 16px rgba(0,0,0,.4); --shadow-lg: 0 8px 32px rgba(0,0,0,.5);
--radius-sm: 6px; --radius-md: 12px; --radius-lg: 20px; --transition: .3s cubic-bezier(.4,0,.2,1);
} /* === 2. RESET === */ *, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }
html { scroll-behavior: smooth; } body { font-family: var(--font-body); background: var(--clr-bg);
color: var(--clr-text); line-height: 1.6; } img { max-width: 100%; height: auto; display: block; }
a { color: var(--clr-primary); } /* === 3. HERO SECTION === */ .hero { min-height: 100svh; display:
grid; place-items: center; background: radial-gradient(ellipse at center, rgba(228,77,38,.15) 0%,
transparent 70%); text-align: center; padding: var(--sp-lg) var(--sp-sm); } .hero__name {
font-size: var(--fs-2xl); font-family: var(--font-heading); line-height: 1; } .hero__role {
font-size: var(--fs-lg); color: var(--clr-primary); } /* === 4. NAVIGATION === */ .site-header {
position: sticky; top: 0; z-index: 100; display: flex; justify-content: space-between; align-items:
center; padding: var(--sp-xs) var(--sp-md); background: rgba(15,15,15,.9); backdrop-filter:
blur(12px); border-bottom: 1px solid rgba(255,255,255,.08); } .nav__list { display: flex; gap:
var(--sp-md); list-style: none; } .nav__list a { text-decoration: none; color:
var(--clr-text-muted); transition: color var(--transition); } .nav__list a:hover { color:
var(--clr-primary); } /* === 5. PROJECTS GRID === */ .projects__grid { display: grid;
grid-template-columns: repeat(auto-fill, minmax(min(320px,100%), 1fr)); gap: var(--sp-md); }
.project-card { background: var(--clr-surface); border-radius: var(--radius-md); overflow: hidden;
transition: transform var(--transition), box-shadow var(--transition); } .project-card:hover {
transform: translateY(-8px); box-shadow: var(--shadow-lg); } /* === 6. BUTTONS === */ .btn {

```

```
display: inline-flex; align-items: center; gap: 8px; padding: 12px 28px; border-radius:
var(--radius-sm); font-weight: 600; font-size: var(--fs-sm); text-decoration: none; cursor:
pointer; border: 2px solid transparent; transition: background var(--transition), transform
var(--transition); } .btn:active { transform: scale(.97); } .btn--primary { background:
var(--clr-primary); color: white; } .btn--primary:hover { background: var(--clr-primary-dark); }
.btn--outline { border-color: var(--clr-primary); color: var(--clr-primary); } .btn--outline:hover
{ background: var(--clr-primary); color: white; } /* === 7. RESPONSIVE === */ @media (max-width:
768px) { .nav__list { display: none; } .nav__list.open { display: flex; flex-direction: column; }
.about__grid { grid-template-columns: 1fr; } }
```

SECTION 4

Deploying with GitHub Pages

1	Create Repository GitHub → New repository → Name: username.github.io (for user site) or any name (for project site)
2	Initialize Git git init → git add . → git commit -m 'Initial portfolio build'
3	Push to GitHub git remote add origin https://github.com/username/repo.git → git push -u origin main
4	Enable Pages Repository Settings → Pages → Source: Deploy from branch → Branch: main → folder: / (root) → Save
5	Access Your Site Live at: https://username.github.io or https://username.github.io/repo-name in 2-5 minutes
6	Custom Domain Buy yourname.dev on Porkbun/Namecheap (~\$900/yr) → Add CNAME record → GitHub Pages → Custom domain
7	HTTPS GitHub Pages provides free SSL/HTTPS certificate automatically
8	Auto Deploy Every git push to main automatically redeploys — changes live in 2 minutes

SECTION 5

Portfolio Checklist

Category	Must Have
Content	Professional photo, compelling 1-sentence bio, 3-5 best projects with screenshots, contact form
Technical	Valid HTML (validator.w3.org), no console errors, all links work, images have alt text
Performance	Lighthouse score > 90 all categories, images optimised, no render-blocking resources
Responsive	Works on 320px mobile, 768px tablet, 1440px desktop — test on real devices

Accessibility	Keyboard navigable, focus visible on all interactive elements, colour contrast 4.5:1+
SEO	meta description, og:image, meaningful title, semantic headings h1 > h2 > h3
Projects	Each project: what it does, technologies used, what problem it solves, live link + code link
Polish	Consistent spacing, smooth hover states, loading animations, page transitions

■ PRACTICE Q & A

Q1. What should every project card on your portfolio show?

A: Screenshot/preview image, project title, 1-2 sentence description of what problem it solves, technologies used as tags, link to live demo AND link to GitHub code. The problem-solving angle is what impresses hiring managers.

Q2. Why use Formspree for the contact form instead of a backend?

A: Formspree handles form submission, spam filtering, and email delivery for free — no server, no PHP, no Node.js needed. Just point action='https://formspree.io/f/YOUR_ID' and it works.

Q3. What Lighthouse score should your portfolio achieve?

A: Aim for 90+ on all four categories: Performance, Accessibility, Best Practices, and SEO. Run it in Chrome DevTools (F12 → Lighthouse tab). Fix every red and orange issue — it demonstrates you care about quality.

Q4. What is a skip link and why does every site need one?

A: A visually hidden link at the top of the page that becomes visible on focus, allowing keyboard users to skip past the navigation directly to main content. Without it, keyboard users must tab through every nav link on every page.

Q5. How do you keep your portfolio updated without losing visitors?

A: Keep the URL permanent, update projects in place rather than changing URLs, add new projects at the top, and remove old/irrelevant ones. Update your featured project every 6 months to reflect your current skill level.

■ Portfolio Project Cheat Sheet	
Wireframe first	Plan before coding – always
:root { --var: val; }	Design tokens – change once, updates all
Semantic HTML sections	header, main, section, article, footer
mobile-first + min-width	Start mobile, enhance for desktop
auto-fill minmax(320px, 1fr)	Responsive project grid
skip-link + aria-labels	Essential accessibility
loading='lazy'	Lazy-load project screenshots
og:image meta tag	Social sharing preview
Formspree.io	Free contact form – no backend
GitHub Pages	Free hosting – push to deploy
Lighthouse score > 90	Proves you ship quality code
validator.w3.org	Validate HTML before launch